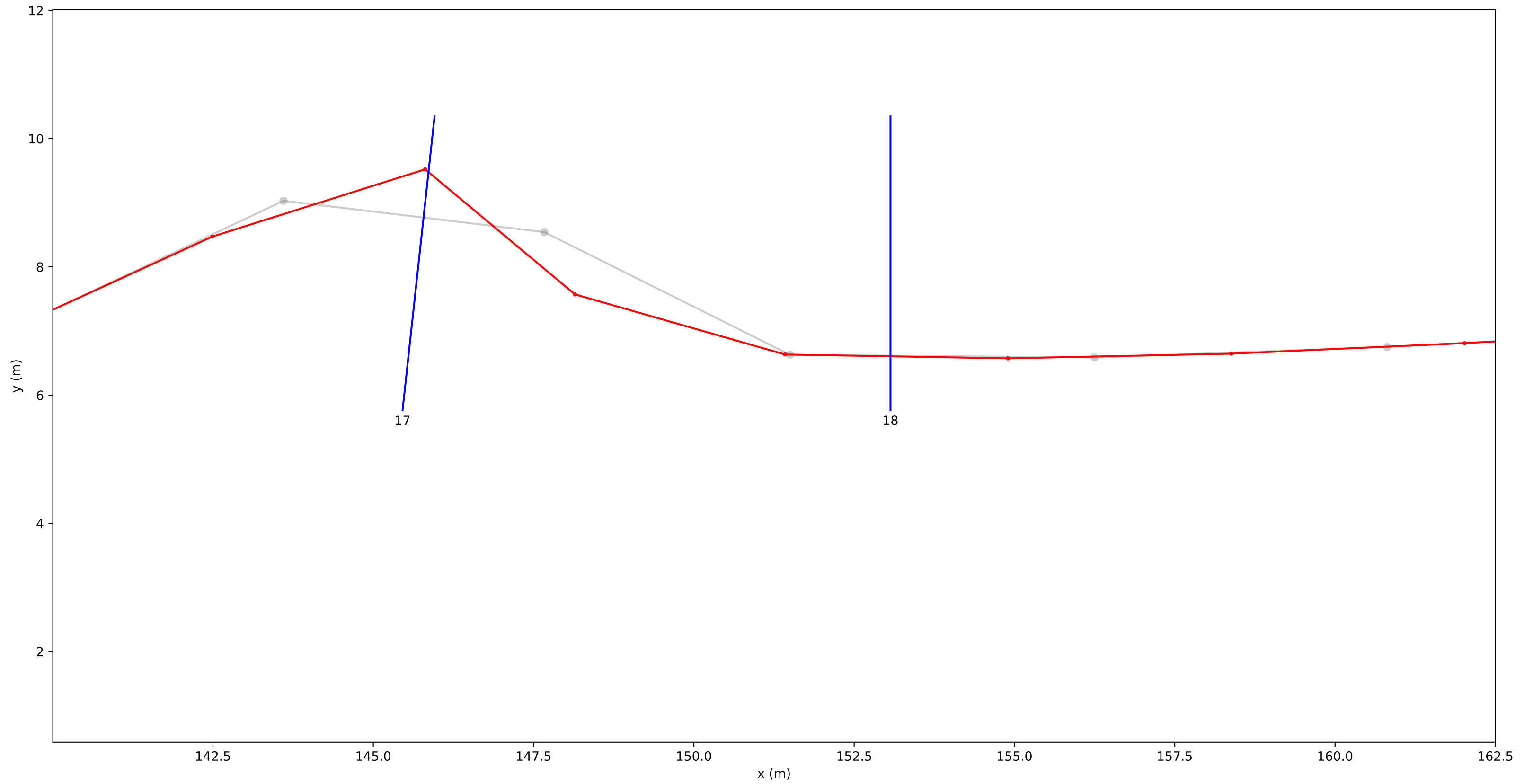


Map plot: optimized path through gates



## Track Generation Report

=====

Generated: 2026-06-01 00:46:41

Gate file: ref\_gates\_endurance.csv

Gate count: 196

Saved DOFs file: opt\_dofs\_2026-06-01\_00-46-41.npy

### Optimization parameters

method: L-BFGS-B

maxiter: 5

ftol: 1e-07

### Point counts

objective function points: 784

path generation points: 588

### Results

total optimization time: 791.660 s

total path length: 2767.2909 m

**total objective function value: 300.745365**

Per-step time estimate (one measured objective evaluation)

make spline from dofs: 34.81 ms

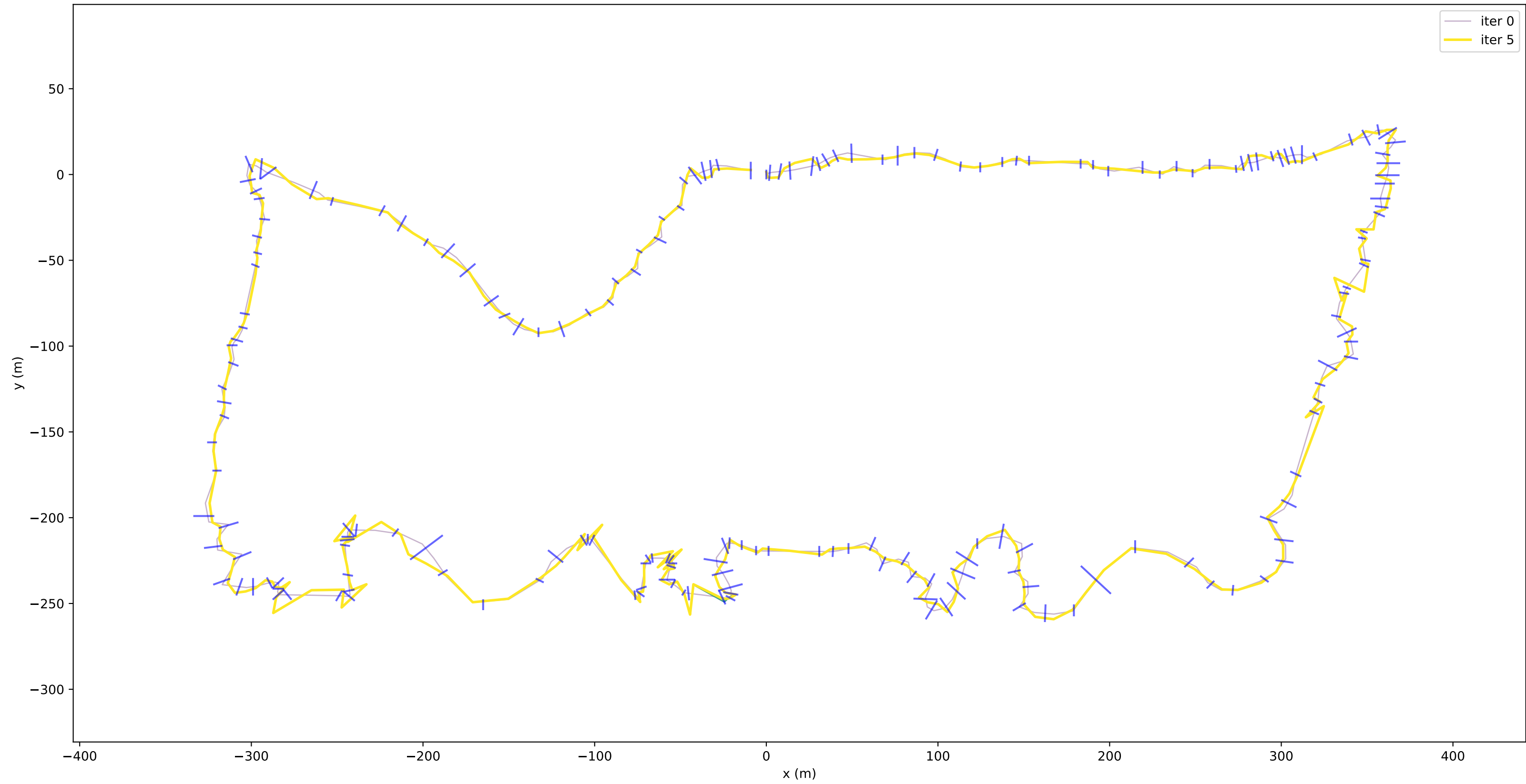
path length function: 0.52 ms

equidistant times generated: 0.53 ms

curvature objective function: 141.23 ms

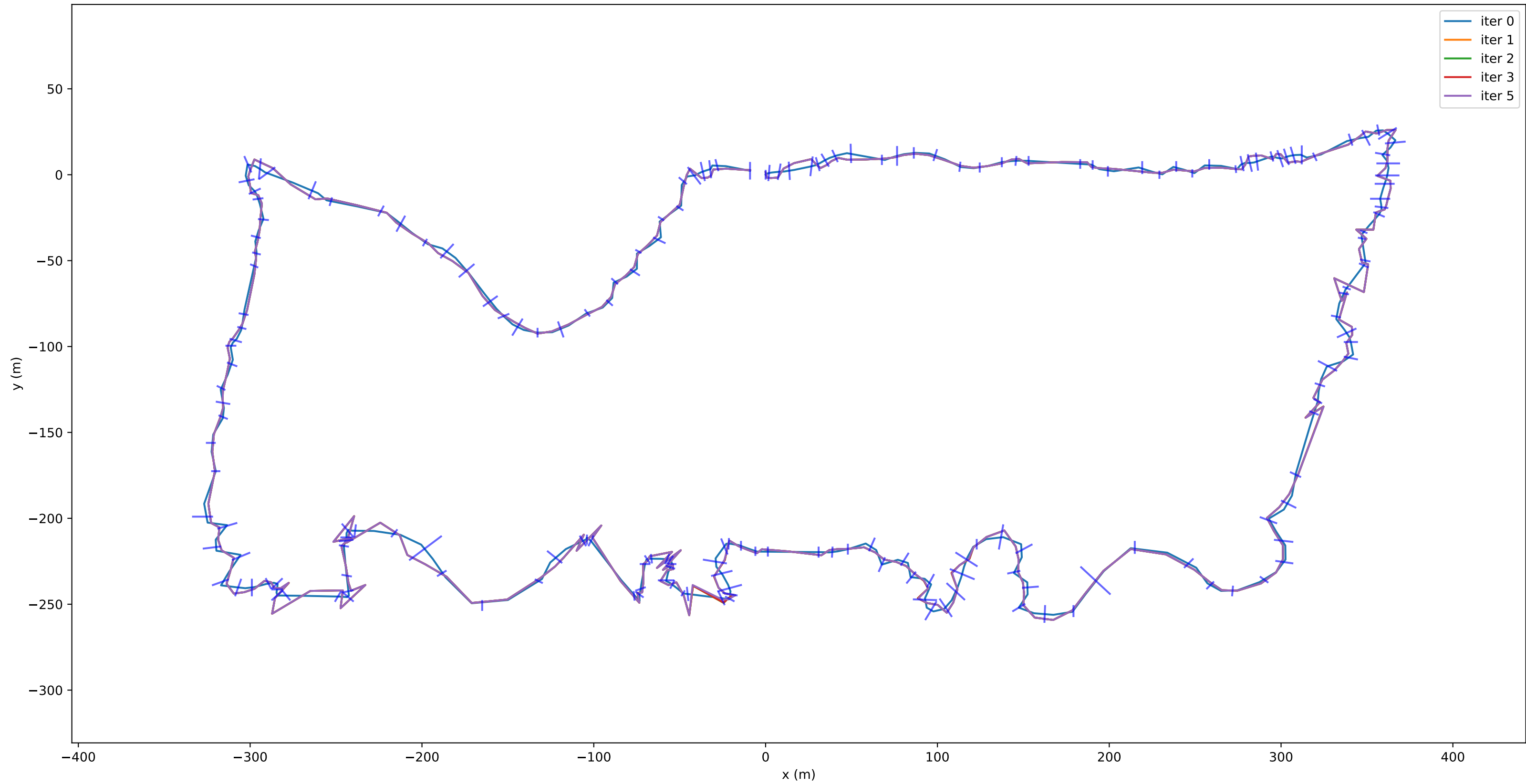
Optimization progression

Optimization progression (6 iterations)



Equally spaced iterations

5 equally spaced iterations



Instant radius vs distance

